

Limits

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Epsilon-Delta Definition of Limits

We say that $f(x)$ approaches the limit L as x approaches x_0 and we write

$$\lim_{x \rightarrow x_0} f(x) = L$$

if given any $\epsilon > 0$ there exists $\delta > 0$ such that

$$0 < |x - x_0| < \delta \rightarrow |f(x) - L| < \epsilon$$

(for examples see written notes)

Triangle Inequality

One has $|x + y| \leq |x| + |y|$ for all x, y

For real numbers this holds because

$$\begin{aligned} |x + y|^2 &= (x + y)^2 \\ &= x^2 + y^2 + 2xy \\ &\leq |x| + |y| + 2|x||y| = (|x| + |y|)^2 \end{aligned}$$

Theorem - Properties of limits

The limit of a sum/product/quotient is the sum/product/quotient of the limits

Namely, if

$$\lim_{x \rightarrow x_0} f(x) = L, \quad \lim_{x \rightarrow x_0} g(x) = M$$

Then

$$\lim_{x \rightarrow x_0} [f(x) + g(x)] = L + M$$

$$\lim_{x \rightarrow x_0} [f(x) \cdot g(x)] = L \cdot M$$

$$\lim_{x \rightarrow x_0} \left[\frac{f(x)}{g(x)} \right] = \frac{L}{M}, \quad M \neq 0$$

Proofs

We use the $\epsilon - \delta$ definition

We know $\lim_{x \rightarrow x_0} f(x) = L$ and $\lim_{x \rightarrow x_0} g(x) = M$

To compute the limit of the sums

We need $\lim_{x \rightarrow x_0} [f(x) + g(x)] = L + M$

Now, let $\epsilon > 0$ be given, we estimate

$$\begin{aligned} |f(x) + g(x) - (L + M)| &= |f(x) - L + g(x) - M| \\ &\leq |f(x) - L| + |g(x) - M| \end{aligned}$$

Since $\lim_{x \rightarrow x_0} f(x) = L$ there exists $\delta_1 > 0$

$$0 < |x - x_0| < \delta_1 \Rightarrow |f(x) - L| < \frac{\epsilon}{2}$$

Since $\lim_{x \rightarrow x_0} g(x) = M$ there exists $\delta_2 > 0$

$$0 \neq |x - x_0| < \delta_2 \Rightarrow |g(x) - M| < \frac{\epsilon}{2}$$

Thus,

$$0 \neq |x - x_0| < \min\{\delta_1, \delta_2\} \Rightarrow |f(x) + g(x) - L - M| < \epsilon$$

To compute the limit of the products

We estimate,

$$\begin{aligned} |f(x) \cdot g(x) - LM| &= |f(x) \cdot g(x) - L \cdot g(x) + L \cdot g(x) - LM| \\ &\leq |g(x)| |f(x) - L| + |L| |g(x) - M| \end{aligned}$$